

Figure S1 Covariate balance for propensity score matching

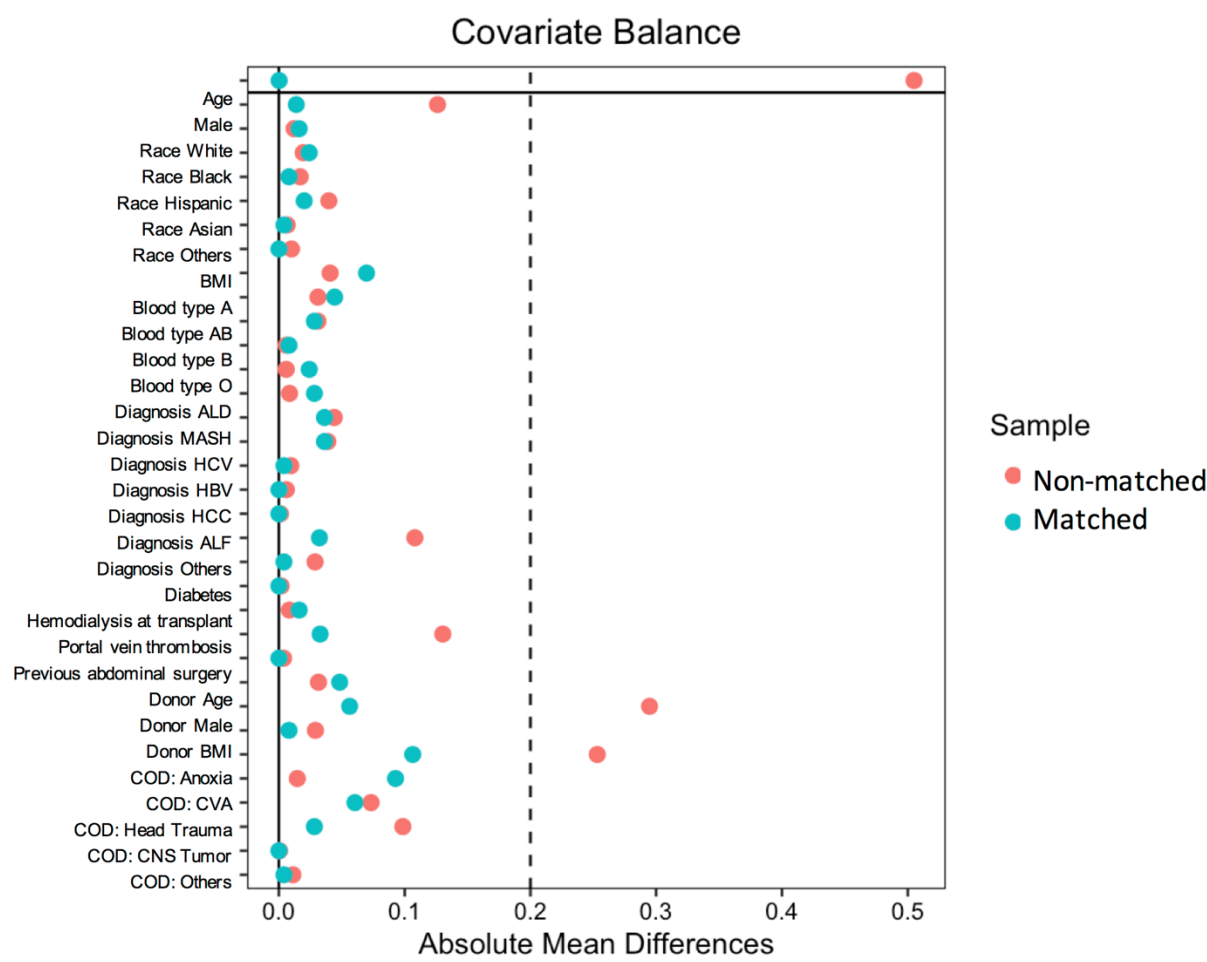


Figure S2 Distribution of propensity score in non-matched and matched cohorts

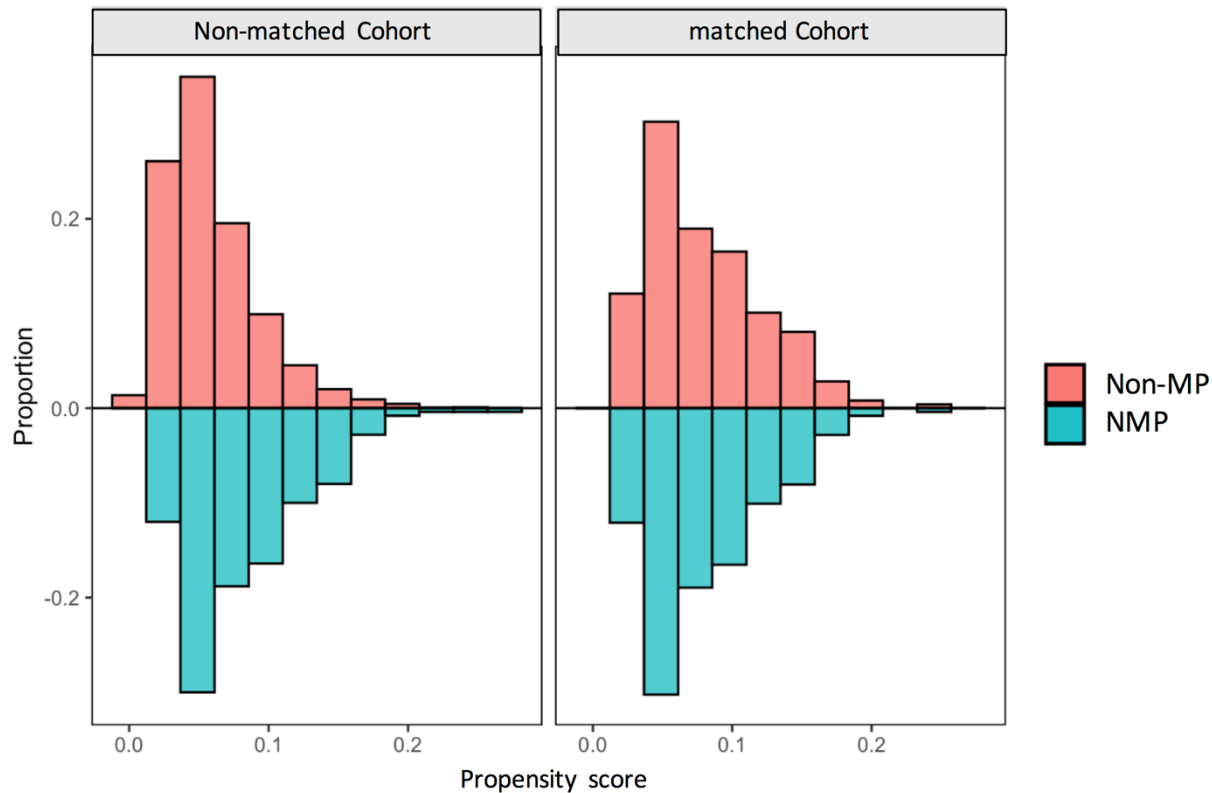


Table S1. Missing data of recipient and donor characteristics, and the outcomes

	Valid N = 4217	Missing N (%)
<Recipient>		
Age	4217	0 (0%)
Gender	4217	0 (0%)
Race	4217	0 (0%)
Body mass index	4216	1 (0.02%)
Blood Type	4217	0
Diabetes	4217	0
Primary Diagnosis of liver disease	4217	0
HCV positive status, N (%)	4217	0
HCV NAT positive, N (%)	4217	0
Previous Abdominal Surgery N (%)	4216	1 (0.02%)
Portal Vein Thrombosis at transplant, N (%)	4216	1 (0.02%)
Dialysis at transplant, N (%)	4183	34 (0.8%)
TIPSS, N (%)	4217	0
MELD at transplant, median (IQR)	4214	3 (0.07%)
MELD exception, N (%)	3764	453 (10.7%)
Mechanical Ventilation at transplant, N (%)	4217	0
Waiting days, median (IQR)	4217	0
<Donor>		
Age	4217	0
Gender	4217	0
Race	4217	0
Blood Type	4217	0
Body mass index	4213	4 (0.09%)
HCV NAT positive	4217	0
HCV Ab positive	4216	1 (0.02%)
Causes of death	4217	0
Share Type	4217	0
Donor Warm Ischemia Time	4204	13 (0.3%)
Preservation Time	4205	12 (0.3%)
DRI	4204	13 (0.3%)
Distance	4217	0
Liver machine perfusion	4217	0
<Outcome>		
Patient status	4217	0
Patient time	4217	0
Graft status kidney	4217	0
Graft time kidney	4217	0
Length of hospital stay	4175	42(1.0%)
Rejection episode before discharge	4215	2 (0.05%)

Rejection episodes at 6month	4217	0
Rejection episodes at 1y	4217	0

Table S2. Demographics of transplant between liver machine perfusion and non-machine perfusion in matched cohort

	NMP N = 248	Non-MP N = 248	P
<Recipient>			
Age, year-old median (IQR)	61.0 (54.0-66.0)	61.0 (54.0-66.0)	0.70
Male, N (%)	169 (68%)	165 (67%)	0.70
Race, N (%), White	187 (75%)	181 (73%)	0.96
Black	8 (3.2%)	10 (4.0%)	
Hispanic	42 (17%)	47 (19%)	
Asian	9 (3.6%)	8 (3.2%)	
Others	2 (0.8%)	2 (0.8%)	
Body mass index, kg/m2, median (IQR)	28.2 (24.9-32.8)	28.2 (24.9-32.0)	0.82
Blood Type, N (%) O	130 (52%)	141 (57%)	0.15
B	20 (8.1%)	22 (8.9%)	
A	91 (37%)	84 (34%)	
AB	7 (2.8%)	1 (0.4%)	
Diabetes, N (%)	85 (34%)	86 (35%)	0.92
Primary Diagnosis of liver disease, N (%)			
ALD	82 (33%)	75 (30%)	0.41
MASH	48 (19%)	57 (23%)	
Hepatitis C virus	17 (6.9%)	8 (3.2%)	
Hepatitis B virus	2 (0.8%)	1 (0.4%)	
Hepatocellular carcinoma	14 (5.6%)	14 (5.6%)	
OTHERS	85 (34%)	93 (38%)	
HCV positive status, N (%)	38 (15%)	40 (16%)	0.81
HCV NAT positive, N (%)	8 (4.4%)	7 (7.1%)	0.34
Previous Abdominal Surgery N (%)	128 (52%)	140 (56%)	0.28
Portal Vein Thrombosis at transplant, N (%)	36 (15%)	40 (16%)	0.62
Dialysis at transplant, N (%)	4 (1.6%)	4 (1.6%)	0.99
TIPSS, N (%)	17 (6.9%)	21 (8.5%)	0.50
MELD at transplant, median (IQR)	18 (11-23)	17 (12-23)	0.94
MELD exception, N (%)	43 (17%)	60 (24%)	0.077
Mechanical Ventilation at transplant, N (%)	1 (0.4%)	1 (0.4%)	0.99
Waiting days, median (IQR)	119 (33-338)	98 (28-248)	0.11
<Donor>			
Age, year-old median (IQR)	43.0 (32.0-52.0)	44.0 (34.0-52.0)	0.51
Male, N (%)	175 (71%)	173 (70%)	0.84
Race, N (%) White	183 (74%)	166 (67%)	0.20
Black	25 (10%)	37 (15%)	
Hispanic	31 (12%)	35 (14%)	
Asian	4 (1.6%)	8 (3.2%)	
Others	5 (2.0%)	2 (0.8%)	

Blood Type, N (%) O	128 (38%)	144 (58%)	0.090
B	22 (8.9%)	20 (8.1%)	
A	93 (38%)	84 (34%)	
AB	5 (2.0%)	0 (0%)	
Body mass index, kg/m2, median (IQR)	28.1 (24.7-33.1)	27.2 (23.9-32.1)	0.21
HCV NAT positive, N (%)	8 (3.2%)	6 (2.4%)	0.59
HCV Ab positive, N (%)	22 (8.9%)	16 (6.5%)	0.31
Causes of death			0.22
Anoxia	137 (55%)	114 (46%)	
CVA	61 (25%)	76 (31%)	
Head Trauma	39 (17%)	46 (19%)	
Others	11 (4.4%)	12 (4.8%)	
Share Type, N (%) Local	134 (54%)	151 (61%)	0.003
Regional	60 (24%)	71 (29%)	
National	54 (22%)	26 (10%)	
Donor WIT, minutes, median (IQR)	23 (19-26)	22 (18-26)	0.42
Preservation Time, hours, median (IQR)	10.1 (7.7-13.3)	5.3 (4.5-6.4)	<0.001
DRI, median (IQR)	2.35 (2.04-2.80)	2.30 (1.99-2.64)	0.042
Distance ^s , median (IQR)	92 (13-270)	60 (9-218)	0.032

IQR: interquartile range, ALD: alcohol related liver disease, MASH: metabolic dysfunction-associated steatohepatitis, DCD: donation after cardiac death, DBD: donation after brain death, TIPSS: transjugular intrahepatic portosystemic shunt, HCV: hepatitis C virus, MELD: model for end stage liver disease, WIT: warm ischemia time, DRI: donor risk index, Distance^s: distance between donor and recipient hospital

Table S3. Univariable Cox-Hazard regression analysis of liver transplant recipient for overall survival and graft survival in non-matched cohort and matched cohort

Non-matched cohort						
	Overall survival			Graft survival		
	Hazard Ratio	95% CI	P	Hazard Ratio	95% CI	P
Recipient						
Age	1.03	1.02, 1.05	<0.001	1.01	1.00, 1.02	0.088
Male	1.28	1.01, 1.62	0.038	1.31	1.09, 1.58	0.004
Race White	1.06	0.82, 1.35	0.67	1.02	0.97, 1.07	0.46
Blood Type O	1.05	0.86, 1.29	0.63	0.98	0.83, 1.16	0.83
BMI	1.00	0.98, 1.02	0.78	1.01	1.00, 1.02	0.20
Diabetes	1.18	0.96, 1.47	0.12	1.12	0.95, 1.34	0.18
HCV positive	1.20	0.95, 1.53	0.13	1.02	0.83, 1.24	0.87
Portal vein thrombosis	1.17	0.88, 1.55	0.28	1.20	0.96, 1.50	0.11
Dialysis	2.10	1.18, 3.74	0.011	2.24	1.43, 3.50	<0.001
MELD score	1.01	1.00, 1.02	0.22	1.01	1.00, 1.02	0.11
Previous abdominal surgery	1.06	0.86, 1.30	0.58	1.03	0.87, 1.21	0.73
Mechanical ventilation	1.64	0.61, 4.39	0.33	2.64	1.41, 4.94	0.002
Donor						
NMP	0.51	0.26, 0.98	0.044	0.55	0.33, 0.90	0.018
Age	1.00	1.00, 1.01	0.41	1.01	1.00, 1.02	0.001
Male	1.01	0.81, 1.25	0.96	1.14	0.96, 1.37	0.14
Race White	1.02	0.80, 1.30	0.87	1.06	1.00, 1.12	0.046
BMI	1.00	0.98, 1.01	0.60	1.01	0.99, 1.02	0.37
Causes of death CVA	1.59	1.25, 2.01	<0.001	1.74	1.44, 2.09	<0.001
Preservation time	1.02	0.97, 1.06	0.44	1.03	1.00, 1.06	0.094
Warm ischemia time	1.00	1.00, 1.00	0.15	1.00	1.00, 1.00	0.071
Donor risk index	1.02	0.83, 1.33	0.66	1.27	1.06, 1.52	0.008
Matched cohort						
	Overall survival			Graft survival		
	Hazard Ratio	95% CI	P	Hazard Ratio	95% CI	P
Recipient						
Age	1.02	0.98, 1.06	0.31	1.00	0.98, 1.03	0.78
Male	1.22	0.59, 2.54	0.59	1.15	0.65, 2.02	0.63
Race White	0.72	0.35, 1.47	0.37	0.57	0.33, 0.97	0.039
Blood Type O	1.06	0.54, 2.08	0.86	0.78	0.46, 1.31	0.34
BMI	1.00	0.94, 1.06	0.89	1.00	0.95, 1.05	0.91
Diabetes	1.73	0.89, 3.37	0.11	1.72	1.02, 2.89	0.043
HCV positive	0.84	0.32, 2.15	0.71	0.48	0.19, 1.21	0.12
Portal vein thrombosis	1.74	0.79, 3.83	0.17	1.90	1.04, 3.47	0.037
MELD score	1.00	0.95, 1.04	0.94	1.01	0.97, 1.04	0.65
Previous abdominal surgery	2.14	1.03, 4.46	0.042	0.18	0.70, 1.99	0.54
Mechanical ventilation	-	-	-	5.84	0.81, 42.2	0.081
Donor						
NMP	0.42	0.20, 0.91	0.027	0.42	0.23, 0.77	0.005
Age	1.02	1.00, 1.06	0.10	1.03	1.00, 1.05	0.024
Male	1.53	0.69, 3.37	0.29	1.88	0.97, 3.63	0.060
Race White	1.48	0.67, 3.25	0.33	1.34	0.73, 2.45	0.34
BMI	1.02	0.97, 1.07	0.46	1.01	0.97, 1.05	0.58

Causes of death CVA	2.28	1.17, 4.44	0.015	1.69	0.99, 2.89	0.053
Preservation time	0.97	0.88, 1.06	0.49	0.98	0.91, 1.05	0.51
Warm ischemia time	0.99	0.96, 1.02	0.48	0.99	0.97, 1.01	0.36
Donor risk index	1.50	0.83, 2.70	0.18	1.61	1.03, 2.52	0.038